# Lecture 8 Notes

**Quantitative Data Analysis**

# Introduction

- Today we will learn more of the magic of the linear model
- So far we have looked at the bivariate case i.e., we have analysed two variables at a time
- The power of linear regression comes when we include multiple variables at the same time

# Adding multiple variables

- Adding more variables to our model is as simple as using the + sign
- We start our formula, as always, with the dependent variable followed by the ~ sign
- We then type as many variables as we want in our model separated by the + sign

# Model building

- There are several approaches to model building and variable selection
- Two popular approaches are
    1. Specific to general - start with what we believe is important and add to it
    2. General to specific - start with everything and remove things which aren't significant
- Different researchers and disciplines have different ideas about what is best
- Different approaches may fit different circumstances

# First things to look at

- When looking at the model output I would generally look at:
    1. The *sign* of the coefficients
    2. The statistical significance of the coefficients
    3. The magnitude of the coefficients
    4. The $R^2$ value
- Are they reasonable?
- Let's try with our example

# What to do about insignificant variables

- By the time we got to the end of our model, we had a lot of statistically insignificant variables

- By this, we mean that we cannot rule out the true association between this variable and the dependent variable is zero
- This raises the question: should we throw these variables out of our model?

# The argument for removing

- Some approaches to data analysis advocate throwing the variables out
- There are a variety of automated approaches for achieving this
- If we can't rule out no association, then we can't make any claim to knowledge
- If we can't do that, we should throw it out

# The argument against removing

- As social scientists, we generally take a *deductive* rather than *inductive* approach to data analysis
- We construct our models based on theory and/or prior literature
- If our theory demands that a variable be in the model, then it should be there
- If the theory says we should control for something, then it may be useful to have it there
- If something isn't significant when theory/literature suggests it should be, then this may be a warning to check what we are doing (more on this later)

# Some practical advice

- Sometimes we include some variables where it isn't clear *a priori* whether or not they will be significant
- Perhaps in some studies the variables have been included but with mixed results
- If we didn't have a strong reason to include the variable and it is not significant, then it is probably safe to drop it
- If a variable has the wrong sign and is statistically insignificant, then we may also consider dropping
- There are automated approaches to select variables, but these are usually a terrible idea in a social science context

# An alternative measure

- When we started thinking about our model in the first lecture, we outlines some concepts we thought were important
- We hypothesised that education would be related to wages

- We have included a measure of education in our model
- However, other measures are possible
- If our results are unexpected, we may consider changing our measure
- We may experiment with several measures to check the robustness of our approach

# Factor variables

- We included many factor variables in our regression
- Some of these had many categories; maybe too many
- If many of the levels of our factors are insignificant, we may consider reducing the number of categories
- For example, if we have age split into 20 categories, we may reduce it to 5
- We may also change the reference category e.g., using the smallest industry as a reference level is probably a bad idea

# The $R^2$ value

- The $R^2$ value tells us how much variation in the dependent variable is explained by variation in the independent variables
- Many become fixated on having the value as high as possible
- A good model doesn't have to have a high value
- In judging whether the value you have is reasonable:
    - Are there previous studies to compare to?
    - How important do you think the variable you have included are?

# A problem with $R^2$

- We observed that as we added variables to the model the $R^2$ always increased
- It will never decrease when you add variables, even if they are insignificant
- To get around this, someone clever proposed the adjusted $R^2$
- This measure is much the same as the regular version, but penalises the inclusion of irrelevant variables
- Helpfully, it is reported as part of the regression output

# Goodness of fit

- The $R^2$ and adjusted- $R^2$ are example of measures of goodness of fit

- Such measures attempt to tell us how well the model fits the data
- These are the two most common measures for the regression models we work with
- Other measures exist which can be used with a wider variety of regression models
- Akaike information criterion (AIC)
- Log-likelihood
- Bayesian Information Criterion (BIC) (sometimes called the Schwarz Information criterion or Schwarz-Bayes Information Criterion)

# What next?

- We have now discussed some rules we might use to refine our model
- Unfortunately it's not that simple
- Our interpretation of the model results is only correct if certain assumptions hold
- The more we depart from the assumptions of the model, the less robust are our findings
- First let's talk about the ways we may be wrong
- We will assume that our data are representative of the population of interest

# Bias

- Bias is a word which will strike fear into the heart of a modeller
- A coefficient in our model is biased if the expected value of that estimate is not the population value of the parameter
- Formally, for an estimate of a parameter to be unbiased, the following must be true $E[\hat{\beta}] = \beta$
- If, on average, our estimator doesn't produce the population parameter then we have a problem
- We (potentially) can't trust the direction or size of the effect
- We (potentially) can't say anything meaningful about statistical significance

# Consistency

- An estimator may be unbiased or biased
- It may be consistent or inconsistent
- A biased estimator may be saved if it is consistent
- A consistent estimator will converge on the population parameter if our sample size is sufficiently large

- Formally, an estimator is consistent if $\text{plim}_{n\to\infty}\hat{\beta} = \beta$
- Having enough data might deal with bias if the estimator is consistent
- If it is biased and inconsistent then we have a problem

# Efficiency

- Although not crucial, we generally like our estimator to be as efficient as possible
- The more efficient the estimator, the more precise our estimates will be i.e., the standard errors will be smaller
- If our estimator isn't efficient, we might end up committing a Type II error
- This isn't the most serious problem, but one to bear in mind

# Best Linear Unbiased Estimator (BLUE)

- Under a particular set of assumptions, the OLS regression is BLUE
- It is unbiased i.e. $E[\hat{\beta}] = \beta$
- It has the smallest variance among linear estimators i.e., it is efficient
- When we work with this model, we must be aware of the assumptions

# Quotes

> Assumptions are dangerous things.

*Agatha Christie*

> Assumptions are dangerous things to make, and like all dangerous things to make – bombs, for instance, or strawberry shortcake – if you make even the tiniest mistake you can find yourself in terrible trouble.

*Lemony Snicket, The Austere Academy*

> Euclid taught me that without assumptions there is no proof. Therefore, in any argument, examine the assumptions.

*E.T. Bell*

# Assumptions

We will consider the following assumptions (taken from Peter Kennedy's "A guide to econometrics"):

1. The model is correctly specified
2. The residuals have the same variance and are independent of one another
3. There are no exact linear relationships between independent variables
4. The independent variables are fixed and measured without error
5. The expected value of the residual is zero

# Assumption 1

- If you mess up on this assumption, you risk the rest of the work being useless
- There are many parts to this assumption
- As we are on the topic of variable selection, we will start there

# Omitted variable bias

- In our case, we are interested in whether there is a difference in hourly wage between men and women
- We want to focus in on the idea of discrimination i.e., if we have two people who are identical in every respect apart from sex, will they be paid different amounts?
- We must therefore *control* for anything which:
    1. Determines wages
    2. Varies by sex
- If we don't include this, we generate *omitted variable bias*

# Omitted variable bias

- This is a serious form of bias and affects the interpretation we can give to our model
- If we omit one of these variables, our sex coefficient will capture the effect of the omitted variable to the extent that they are correlated
- We should include any such variables in the regression
- We should identify any that we are unable to control for and adjust our interpretation accordingly
- This can be listed as a limitation of our analysis

# Inclusion of irrelevant variables

- Ideally we don't want to include irrelevant variables in our model

- Including an irrelevant variable decreases the efficiency of our estimation
- It also makes our model harder to present to our reader
- Bias is worse than a loss of efficiency, so if in doubt, leave it in

# Deciding what to include

- Don't forget that this process should be guided by theory and previous work
- Social Scientists generally do not decide on what variables to include in a mechanical fashion
- Always take time to reflect on what you are doing and whether it makes sense
- Never be afraid to return to the literature to see what has already been done

# More on what to include

- I have been talking about what variables to include in our model
- Much earlier on we talked about who would be included in our population and sample
- However, what about outliers i.e., points which are far away from the others?
- Different disciplines and authors have different ideas about how to handle this
- Let's consider a **fictional example** i.e., not our labour force survey data

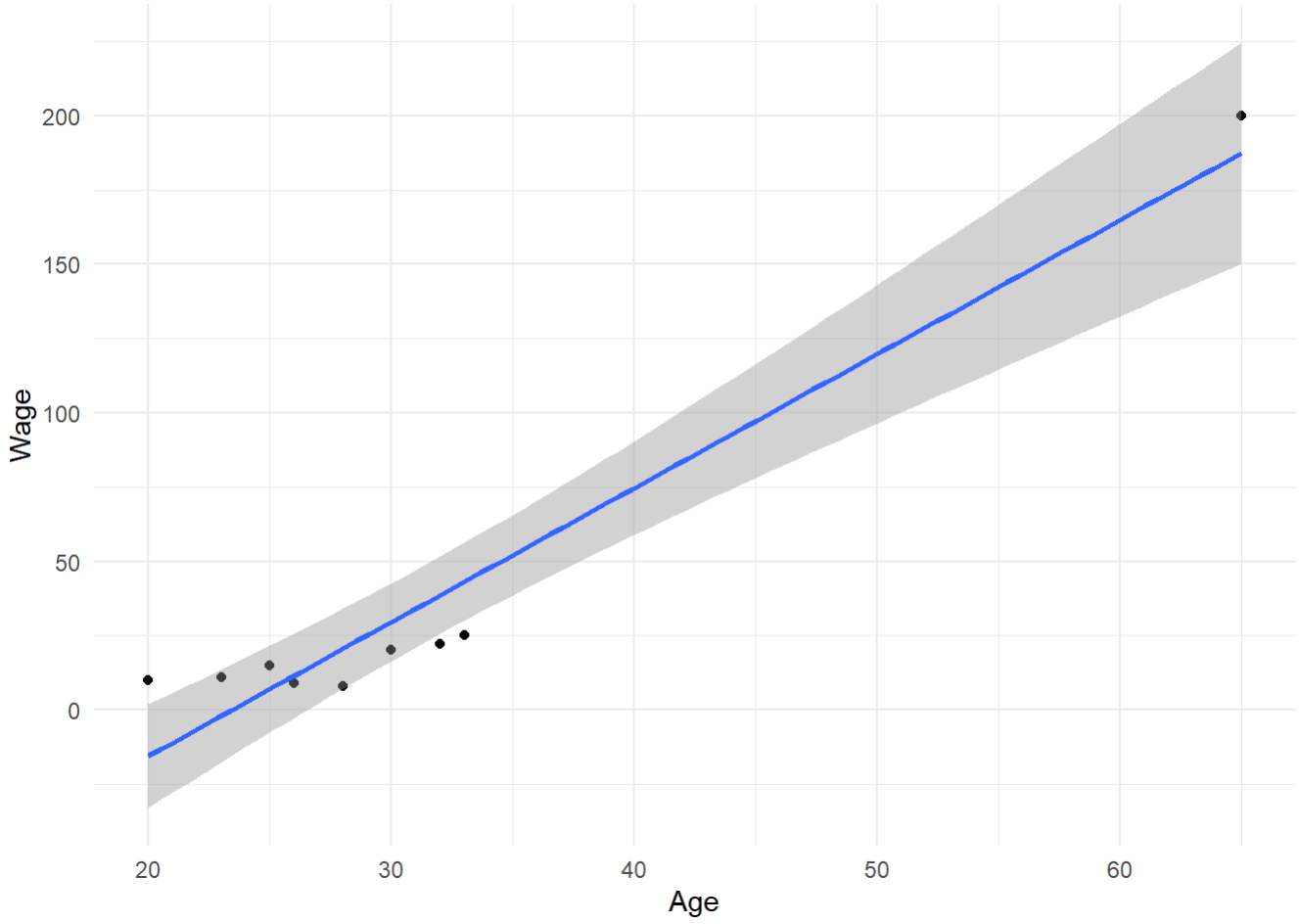
# Making some fake data

```
fake_data <- data.frame(age= c(20, 23, 25, 26, 28, 30, 32, 33, 65 ),
                        wage = c(10, 11, 15, 9, 8, 20, 22, 25, 200))
```

# Wage versus age

```
##
## Call:
## lm(formula = wage ~ age, data = .)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -18.075 -12.517  -2.494  12.556  25.575
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -105.8059    14.7808  -7.158 0.000184 ***
## age            4.5115     0.4379  10.302 1.76e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 16.48 on 7 degrees of freedom
## Multiple R-squared:  0.9381, Adjusted R-squared:  0.9293
## F-statistic: 106.1 on 1 and 7 DF,  p-value: 1.758e-05
```

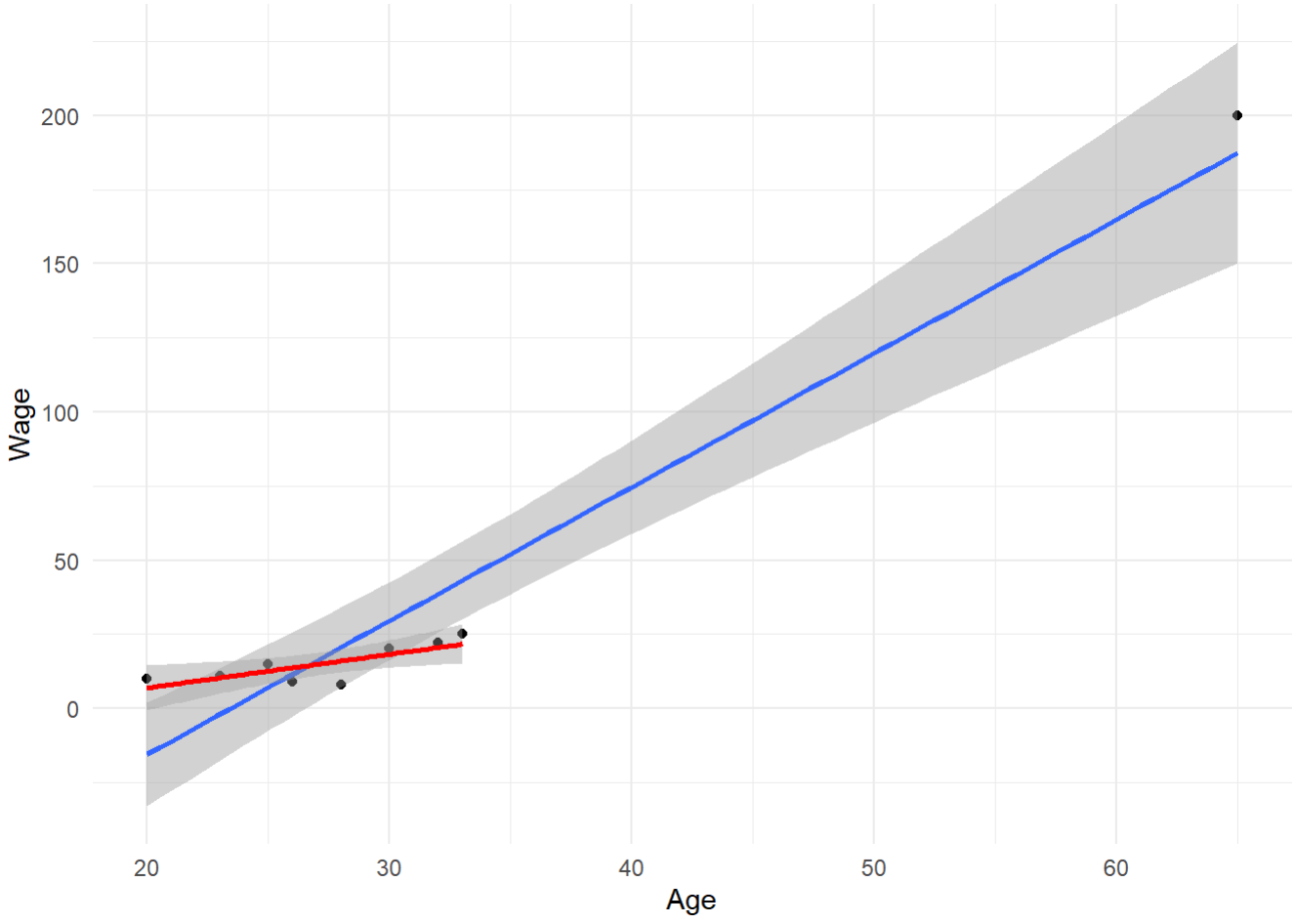
# Wage versus age




# Wage versus age excluding the outlier

```
##
## Call:
## lm(formula = wage ~ age, data = .)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.9938 -0.6668  1.5989  2.5832  3.3274
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -15.8075    10.2675  -1.540   0.1746
## age           1.1358     0.3741   3.036   0.0229 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.44 on 6 degrees of freedom
## Multiple R-squared:  0.6057, Adjusted R-squared:   0.54
## F-statistic: 9.218 on 1 and 6 DF,  p-value: 0.02292
```

# Wage versus age




# Identifying outliers

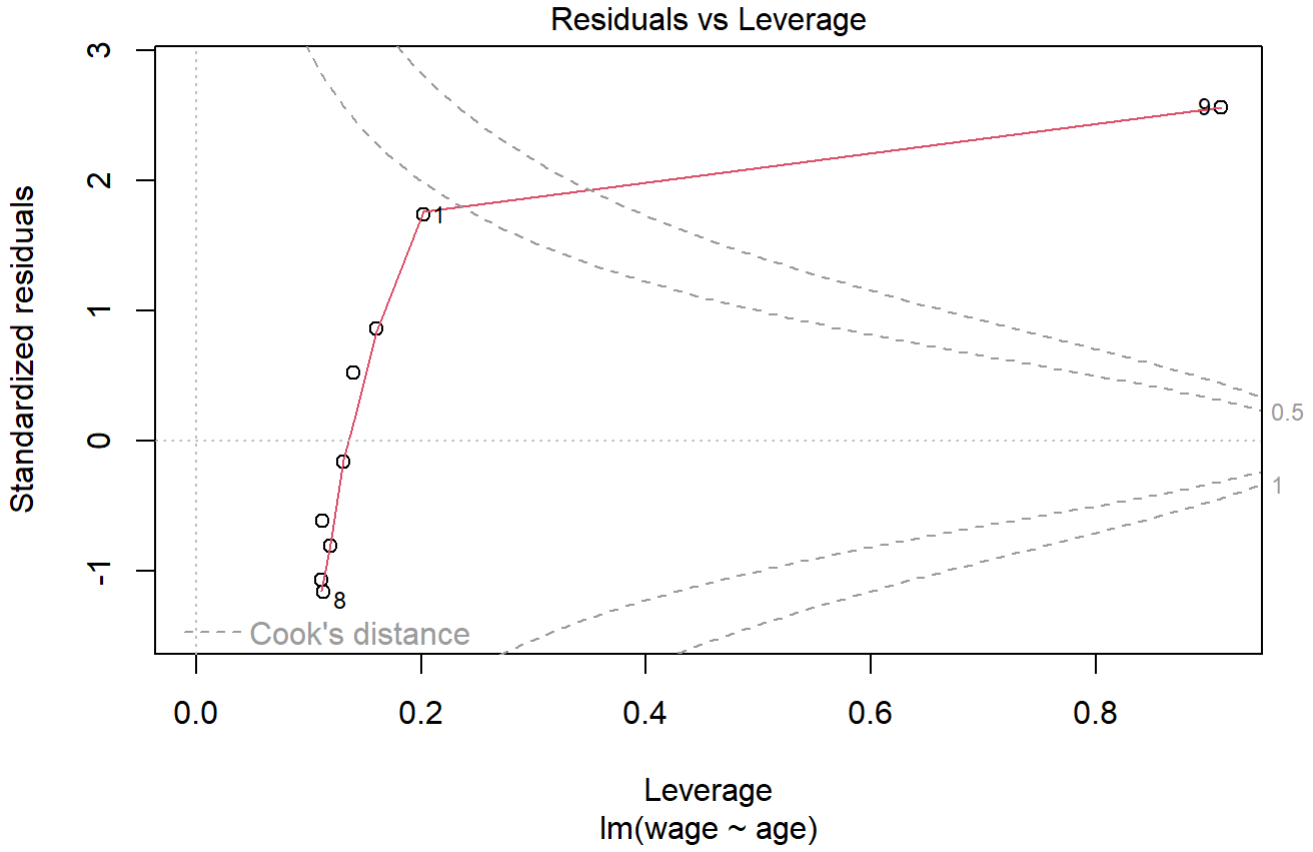
- We can start to identify outliers when describing our data
- Are there any points which look different from the others?
- In the first instance, decide whether the values are feasible
- For example, ages of -10 or 634 are not feasible values
- These should be removed

# What if they are valid values?

- Some disciplines recommend removing outliers
- We have to be careful not to exclude observations just because they don't fit in with our model
- Before worrying about it, we should try to work out whether it makes any difference
- In our fictional example, it clearly matters
- We can look at a residual versus leverage plot to help us figure out how influential a point is

# Residuals vs Leverage

```
fake_data %>% lm(wage ~ age, data=.) %>% plot(which=5)
```

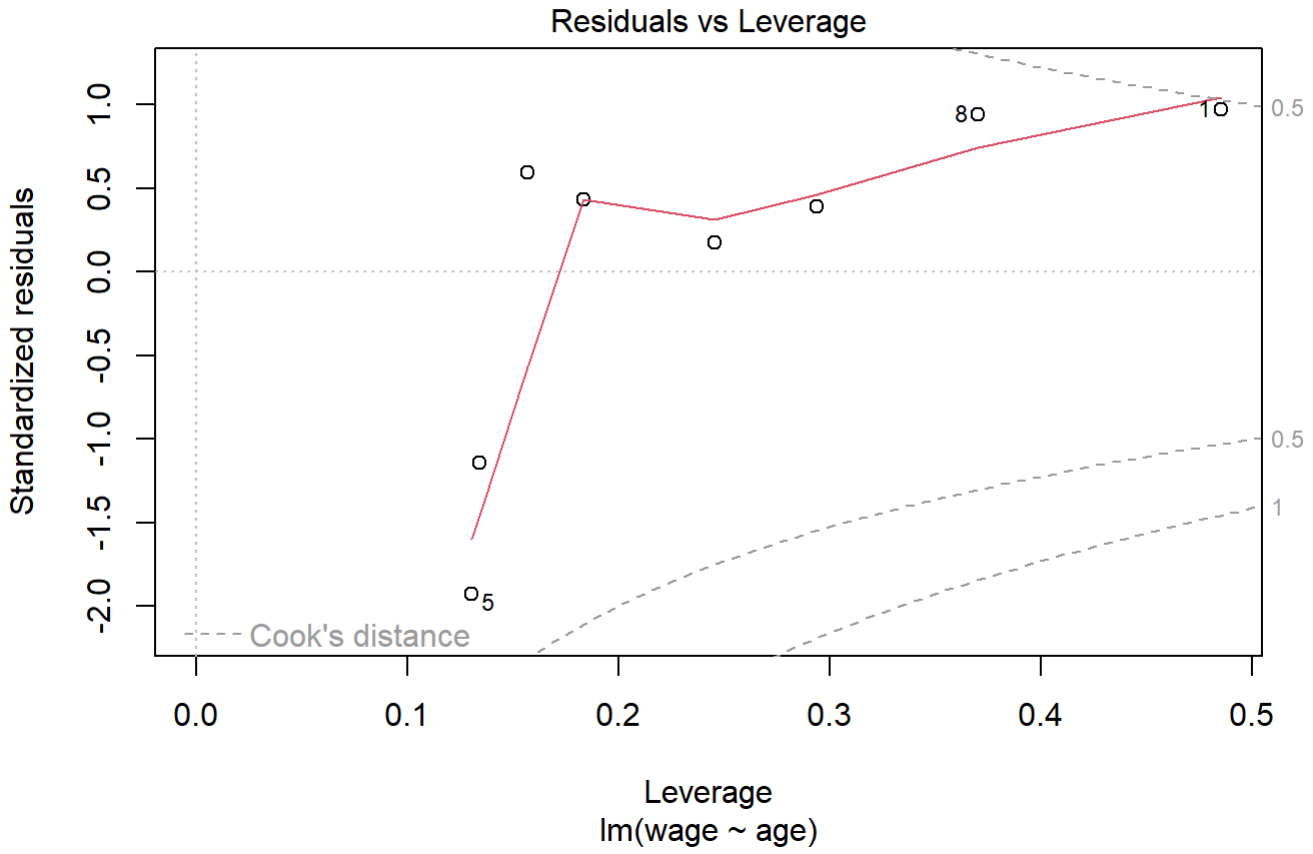


# Residuals vs Leverage

- The plot highlights that observation 9 is highly influential
- Our results look quite different without it
- We therefore have to think carefully about whether to include it
- We might wonder whether it has been properly recorded
- In this case, it is extreme in terms of both our dependent and independent variables
- I might suggest restricting the range of our sample and noting that we cannot necessarily generalise our results to cover older individuals

# Residuals vs Leverage excluding the extreme value

```
fake_data %>%  filter(age<40) %>% lm(wage ~ age, data=.) %>% plot(which=5)
```



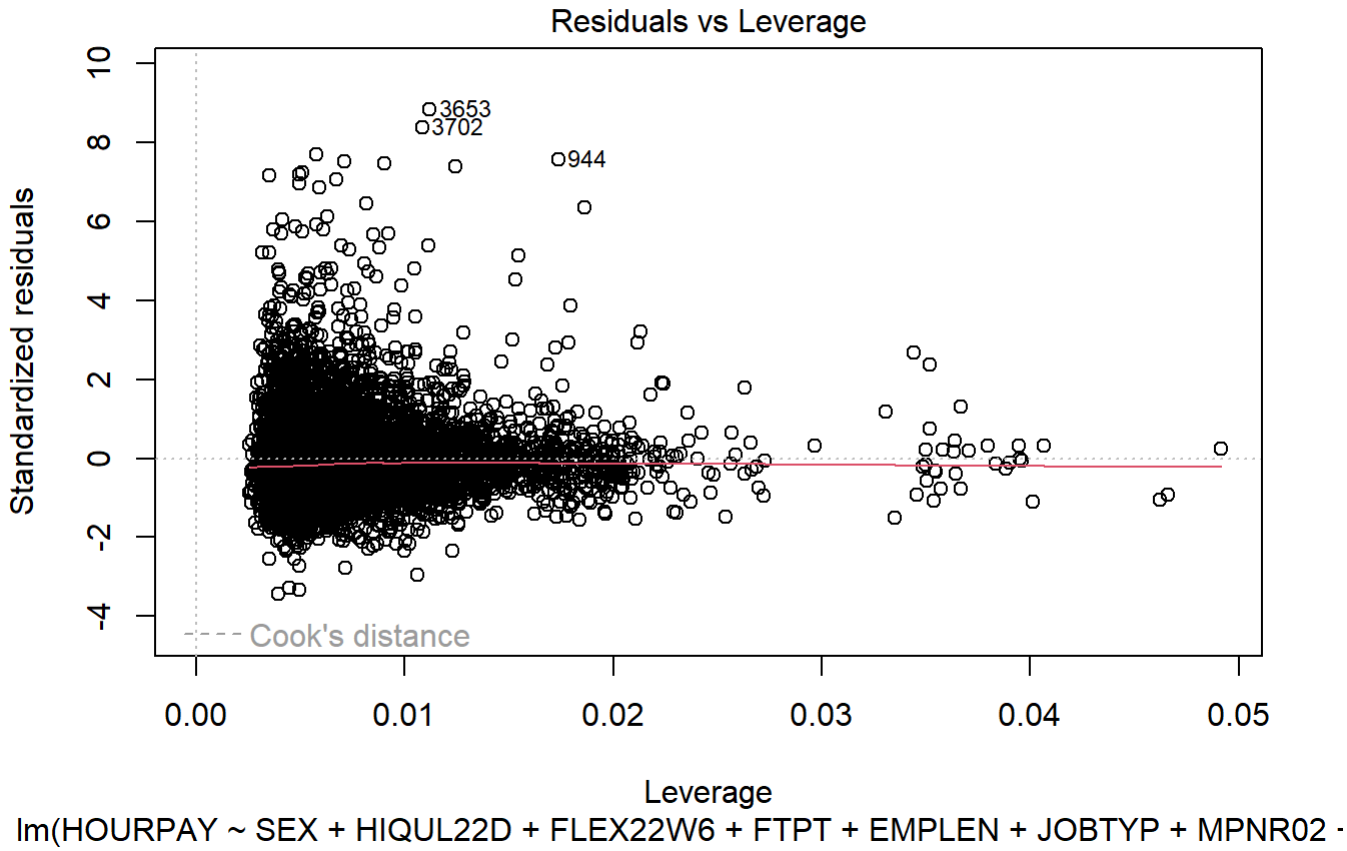

# Thoughts

- People often spend time agonising over outliers when actually they usually don't make any difference

- If you have a large sample size, outliers tend to make very little difference
- Some disciplines suggest applying strict rules in a mechanical fashion to outliers
- Some even use automated approaches to achieve this
- I consider these approaches to be **abominations**
- Always engage your brain and think about the meaning of what you are doing

# Residuals vs Leverage for our regression model

```
lm(HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP + MPNR02 + MARSTA +
INDE07M + NSECMJ20 + AGE + GOR9D, data=lfs) %>% plot(which=5)
```




# Interpretation

- We have some outliers in our data, but they have very little impact on our regression line
- There isn't much need for us to worry about outliers

# Summary

- We have advanced quite a bit in our understanding of linear regression

- We have moved from a bivariate model to a multivariate model
- We have also started to discuss the assumptions
- There are many other assumptions to cover
- We will look at those and how to refine our model